

1. What is a Control System?

বাংলায় বুঝার জন্য:

কোনো সিস্টেমের output-কে desired value অনুযায়ী নিয়ন্ত্রণ করার জন্য input, process এবং feedback-এর সমন্বয়ে যে ব্যবস্থা কাজ করে, সেটাই Control System।

Answer (Write in Exam):

A control system is a system that manages, regulates, or controls the behavior of other devices or systems to achieve a desired output or performance.

2. What is an Open-Loop Control System?

বাংলায় বুঝার জন্য:

যে সিস্টেমে output চেক করে আবার input-এ পাঠানো হয় না, সেটাই Open Loop System।

উদাহরণ: Washing machine, traffic signal.

Answer (Write in Exam):

An open-loop control system is a control system in which the control action is independent of the output. It does not use feedback to control the system.

Example: Washing machine.

3. What is a Closed-Loop Control System?

বাংলায় বুঝার জন্য:

যে সিস্টেমে output measure করে আবার input-এ feedback দেওয়া হয়, সেটাই Closed Loop System।

উদাহরণ: Air conditioner, automatic voltage regulator.

Answer (Write in Exam):

A closed-loop control system is a control system in which the control action depends on the output through a feedback mechanism.

Example: Air conditioner.

4. Differentiate between Open-Loop and Closed-Loop Control Systems.

বাংলায় বুঝার জন্য:

Open Loop-এ feedback নেই, Closed Loop-এ feedback আছে।

Answer (Write in Exam):

Open-Loop System	Closed-Loop System
No feedback is used.	Feedback is used.
Less accurate.	More accurate.
Simple and inexpensive.	Complex and expensive.
Example: Washing machine.	Example: Air conditioner.

5. Draw the Block Diagram of an Open-Loop Control System.

বাংলায় বুঝার জন্য:

Input → Controller → Plant → Output

Answer (Write in Exam):

Input → Controller → Plant → Output

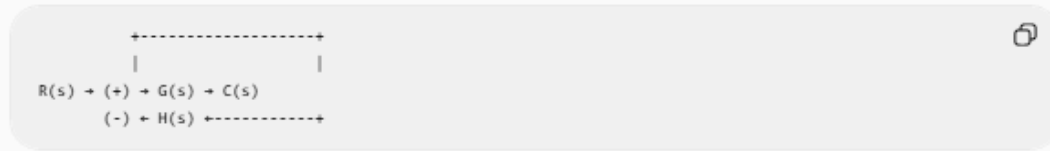


6. Draw the Block Diagram of a Closed-Loop Control System.

বাংলায় বুঝার জন্য:

Output-এর একটা অংশ feedback হয়ে আবার input-এ ফিরে আসে।

Answer (Write in Exam):



Where,

- $R(s)$ = Input
- $G(s)$ = Forward Path Transfer Function
- $H(s)$ = Feedback Transfer Function
- $C(s)$ = Output

7. What is a Transfer Function?

বাংলায় বুঝার জন্য:

Input এবং Output-এর Laplace Transform-এর অনুপাতকে Transfer Function বলে।

Answer (Write in Exam):

The transfer function of a control system is defined as the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming all initial conditions are zero.

$$G(s) = \frac{C(s)}{R(s)}$$

7. What is a Transfer Function?

বাংলায় বুঝার জন্য:

Input এবং Output-এর Laplace Transform-এর অনুপাতকে Transfer Function বলে।

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The transfer function of a control system is defined as the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming all initial conditions are zero.

$$G(s) = \frac{C(s)}{R(s)}$$

8. What are Poles?

বাংলায় বুঝার জন্য:

Transfer Function-এর denominator কে zero করলে যে মানগুলো পাওয়া যায়, সেগুলো হলো poles।

Answer (Write in Exam):

Poles are the values of the complex variable s that make the denominator of the transfer function equal to zero.

Example:

$$G(s) = \frac{10}{s^2 + 5s + 6}$$

Poles:

$$s = -2, -3$$

9. What are Zeros?

বাংলায় বুঝার জন্য:

Transfer Function-এর numerator কে zero করলে যে মানগুলো পাওয়া যায়, সেগুলো হলো zeros।

Answer (Write in Exam):

Zeros are the values of the complex variable s that make the numerator of the transfer function equal to zero.

Example:

$$G(s) = \frac{s + 2}{s^2 + 5s + 6}$$

Zero:

$$s = -2$$

10. What is Feedback?

বাংলায় বুঝার জন্য:

Output-এর একটি অংশকে আবার input-এ ফেরত পাঠানাকে feedback বলে।

Answer (Write in Exam):

Feedback is the process of returning a portion of the output signal to the input of a control system.

11. What is Positive Feedback?

বাংলায় বুঝার জন্য:

যখন feedback signal input-এর সাথে যোগ হয়।

Answer (Write in Exam):

Positive feedback occurs when the feedback signal is added to the input signal, increasing the overall system gain.

12. What is Negative Feedback?

বাংলায় বুঝার জন্য:

যখন feedback signal input থেকে বিয়োগ হয়।

Answer (Write in Exam):

Negative feedback occurs when the feedback signal is subtracted from the input signal, improving system stability and accuracy.

13. What is Block Diagram Reduction?

বাংলায় বুঝার জন্য:

অনেকগুলো block কে step-by-step combine করে একটি equivalent block বের করাকে block diagram reduction বলে।

Answer (Write in Exam):

Block diagram reduction is the process of simplifying a complex control system block diagram into a single equivalent transfer function.

14. What is the Transfer Function of a Closed-Loop System?

বাংলায় বুঝার জন্য:

এই formula টা সবচেয়ে গুরুত্বপূর্ণ।

Answer (Write in Exam):

The transfer function of a closed-loop control system with negative feedback is:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

where,

- $G(s)$ = Forward path transfer function
- $H(s)$ = Feedback transfer function

15. Why is Negative Feedback Preferred in Control Systems?

বাংলায় বুঝার জন্য:

Negative feedback system-কে stable এবং accurate করে।

Answer (Write in Exam):

Negative feedback is preferred because it improves accuracy, increases stability, reduces sensitivity to disturbances, and minimizes errors.

16. Define Stability in a Control System.

বাংলায় বুঝার জন্য:

Input দিলে output যদি নিয়ন্ত্রিত থাকে এবং অসীমে না চলে যায়, তাহলে system stable।

Answer (Write in Exam):

A control system is said to be stable if its output remains bounded for every bounded input.

17. What is a Servo Mechanism?

বাংলায় বুঝার জন্য:

যে closed-loop system mechanical position বা motion control করে তাকে servo mechanism বলে।

Answer (Write in Exam):

A servo mechanism is an automatic closed-loop control system used to control mechanical position, speed, or acceleration.

18. What is Gain in a Control System?

বাংলায় বুঝার জন্য:

Output এবং input-এর amplification ratio।

Answer (Write in Exam):

Gain is the ratio of the output signal amplitude to the input signal amplitude in a control system.

Question 1 — Short Answer (CO1) [$5 \times 1 = 5$ Marks]

(a) ✎: Differentiate between active sensors and passive sensors with suitable examples.

Property	Active sensor	Passive sensor
Power source	Does NOT need external power — generates its own signal	Requires external excitation (power supply)
Output type	Directly generates voltage/current	Changes resistance/capacitance/inductance
Examples	Thermocouple, Piezoelectric sensor, Photovoltaic cell	RTD, Thermistor, Strain gauge, LDR

Key difference: An active sensor like a thermocouple directly produces an EMF (voltage) due to the Seebeck effect without any external power. A passive sensor like an RTD changes its resistance with temperature and needs an external current/voltage source to produce a measurable output.

(b) ✎: State meter loading effect.

The **loading effect** (or meter loading) is the error introduced in a measurement when the measuring instrument draws current from the circuit under test, thereby altering the quantity being measured.

When a voltmeter with internal resistance R_m is connected across a circuit element, it forms a parallel combination with that element. This reduces the effective resistance and thus the measured voltage is lower than the true voltage.

To minimise loading effect → use a voltmeter with very HIGH internal resistance and an ammeter with very LOW internal resistance.

(c) ✎: Define poles and zeros of a transfer function.

Given a transfer function $G(s) = N(s) / D(s)$:

Zeros

Values of s that make the numerator $N(s) = 0$, i.e., $G(s) \rightarrow 0$. These are the roots of $N(s)$.

Poles

Values of s that make the denominator $D(s) = 0$, i.e., $G(s) \rightarrow \infty$. These are the roots of $D(s)$.

Example: $G(s) = (s+2) / [(s+1)(s+3)] \rightarrow$ Zero at $s = -2$; Poles at $s = -1$ and $s = -3$.

(d) **ଉତ୍ତର:** Recall the reasons why RTDs are preferred over thermocouples for precision measurement.

- RTDs have a more linear resistance–temperature relationship compared to the nonlinear EMF–temperature curve of thermocouples.
- RTDs offer higher accuracy and repeatability ($\pm 0.1^\circ\text{C}$ vs $\pm 1\text{--}2^\circ\text{C}$ for thermocouples).
- RTDs have greater stability over long periods; thermocouples drift due to oxidation or wire inhomogeneity.
- RTDs do not require cold junction compensation, unlike thermocouples which need a reference junction.
- RTDs have a higher output signal (resistance change), making them less susceptible to noise interference.

(e) **ଉତ୍ତର:** List two advantages of electronic instruments.

1. **High sensitivity and accuracy:** Electronic instruments can detect and amplify very small signals, giving precise readings that mechanical instruments cannot achieve.
2. **High input impedance:** They draw negligible current from the circuit under test, thereby minimising the loading effect and producing more accurate measurements.

Now Question 2(a) — Current Transformer:

Question 2(a) — Current Transformer (CT) [5 Marks] | CO2

ଉତ୍ତର: Explain the construction and operating principle of a CT. Analyse why the secondary should never be left open while the primary is energized.

Construction

- A toroidal (ring-type) or wound-type iron core provides a closed magnetic path.
- The primary winding consists of very few turns (often just the bus bar passing through the core — $N_1 = 1$). It carries the full line current I_1 .
- The secondary winding has many turns of fine wire ($N_2 \gg N_1$) wound on the core. A standard ammeter (5 A or 1 A range) and a burden resistor Z_B are connected across it.

Operating principle

- The large primary current I_1 creates a magnetic flux in the core.
- This changing flux induces an EMF in the secondary winding by Faraday's law.
- A secondary current I_2 flows through the closed burden circuit such that: $N_1 I_1 \approx N_2 I_2$, so $I_2 = I_1 \times (N_1/N_2)$.
- Since $N_2 \gg N_1$, I_2 is a small, safe, measurable current proportional to I_1 .

⚠ Why the secondary must NEVER be left open while primary is energized

- Normally, the secondary current I_2 produces a counter-MMF ($N_2 I_2$) that opposes the primary MMF ($N_1 I_1$), keeping the net core flux low and the core unsaturated.
- If the secondary is opened, $I_2 = 0$ and there is no opposing MMF. The entire primary MMF now magnetises the core.
- This drives the core into deep saturation, causing the flux to become extremely large.
- By Faraday's law, the rapidly changing saturated flux induces a very high voltage (thousands of volts) across the open secondary terminals — enough to destroy insulation, cause arcing, and be lethal.
- The core also overheats due to excessive eddy current and hysteresis losses.
- **Safety rule:** Always short-circuit the secondary terminals before disconnecting the ammeter or burden.

a) Design the equivalent single transfer function of the control system shown in Figure 3(a) by systematically reducing the block diagram.

[5]

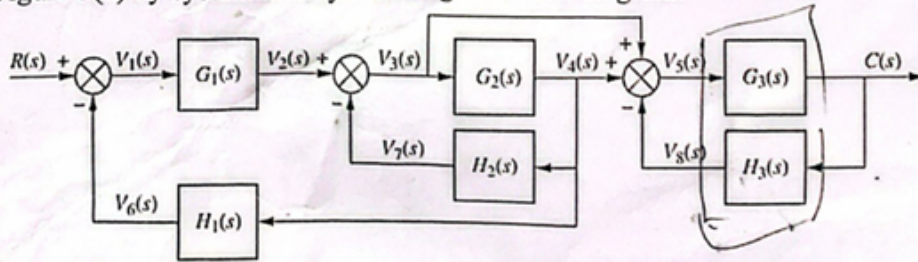


Figure 3 (a)

Step 1: Identify the innermost feedback loop (G_3 - H_3 loop)

ডান পাশে $G_3(s)$ এবং $H_3(s)$ একটি negative feedback loop তৈরি করেছে।

সুতরাং, equivalent transfer function:

$$G_{eq3}(s) = \frac{G_3(s)}{1 + G_3(s)H_3(s)}$$

অর্থাৎ,

$$C(s) = \frac{G_3}{1 + G_3H_3} \times [V_4(s) + V_3(s)]$$

Step 2: Reduce the $G_2 - H_2$ feedback loop

মাঝখানে $G_2(s)$ এবং $H_2(s)$ আরেকটি negative feedback loop।

সুতরাং,

$$G_{eq2}(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$

অর্থাৎ,

$$V_4(s) = \frac{G_2}{1 + G_2H_2} V_2(s)$$

Step 3: Find $V_3(s)$

দ্বিতীয় summing junction থেকে,

$$V_3 = V_2 - V_7$$

এবং,

$$V_7 = H_2 V_4$$

অতএব,

$$V_3 = V_2 - H_2 V_4$$

এখন V_4 বসাই:

$$V_3 = V_2 - H_2 \left(\frac{G_2}{1 + G_2 H_2} \right) V_2$$

$$V_3 = V_2 \left(1 - \frac{G_2 H_2}{1 + G_2 H_2} \right)$$

$$\boxed{V_3 = \frac{V_2}{1 + G_2 H_2}}$$

Step 4: Reduce the outer feedback loop with G_1 and H_1

প্রথম summing junction থেকে,

$$V_1 = R - V_6$$

এবং,

$$V_6 = H_1 V_4$$

যেহেতু,

$$V_2 = G_1 V_1$$

তাহলে,

$$V_2 = G_1 (R - H_1 V_4)$$

এখন V_4 বসাই:

$$V_2 = G_1 \left(R - H_1 \frac{G_2}{1 + G_2 H_2} V_2 \right)$$

সব V_2 একপাশে আনলে,

$$V_2 \left[1 + \frac{G_1 H_1 G_2}{1 + G_2 H_2} \right] = G_1 R$$

অতএব,

$$V_2 = \frac{G_1 (1 + G_2 H_2)}{1 + G_2 H_2 + G_1 H_1 G_2} R$$

Step 5: Find the input of the G_3 block

তৃতীয় summing junction থেকে,

$$V_5 = V_4 + V_3$$

এখন V_4 এবং V_3 বসাই:

$$V_5 = \frac{G_2}{1 + G_2 H_2} V_2 + \frac{1}{1 + G_2 H_2} V_2$$
$$V_5 = \frac{1 + G_2}{1 + G_2 H_2} V_2$$

Step 6: Calculate $C(s)$

যেহেতু,

$$C = \frac{G_3}{1 + G_3 H_3} V_5$$

তাই,

$$C = \frac{G_3}{1 + G_3 H_3} \times \frac{1 + G_2}{1 + G_2 H_2} \times V_2$$

এখন V_2 বসাই:

$$C = \frac{G_3}{1 + G_3 H_3} \times \frac{1 + G_2}{1 + G_2 H_2} \times \frac{G_1(1 + G_2 H_2)}{1 + G_2 H_2 + G_1 H_1 G_2} R$$

এখানে $(1 + G_2 H_2)$ cancel হবে।

Final Equivalent Transfer Function

$$\frac{C(s)}{R(s)} = \frac{G_1(s) [1 + G_2(s)] G_3(s)}{[1 + G_3(s)H_3(s)] [1 + G_2(s)H_2(s) + G_1(s)H_1(s)G_2(s)]}$$

Exam Copy-Paste Final Answer:

$$\frac{C(s)}{R(s)} = \frac{G_1(s) [1 + G_2(s)] G_3(s)}{(1 + G_3(s)H_3(s)) (1 + G_2(s)H_2(s) + G_1(s)H_1(s)G_2(s))}$$